

VOIDSCOPE · BUILD PLAN

VoidScope prototype build plan

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	VDS-BLD-001	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/voidscope		
Source	docs/05-build-plan.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	First build plan; design made constructable (VDS-DDR-002)

VoidScope prototype build plan

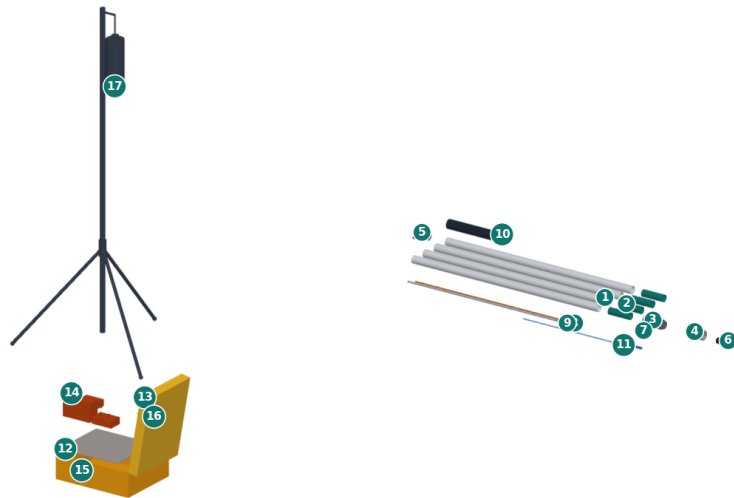
Plan, not yet built. How to build the first proof-of-concept prototype, component by component. Building and testing to it is TRL 4 work. Decisions are kept in the design decisions register ([docs/06-design-decisions.md](#)), not here.

1. What you are building

VoidScope prototype: every component, pulled apart

Numbered in build order. Rods side by side as stored; cable and tubes drawn cut short. Seen from the front right and above

- 1  Rod sections (4), side by side
- 2  Couplers with snap buttons and pins (4)
- 3  Collar
- 4  Nose
- 5  Tail cap, air barb, grommets
- 6  Camera head and spacer ring
- 7  Speaker and cover
- 8  Camera cable (7 m, shown cut)
- 9  Water guide tube (6.5 m, shown cut)
- 10  Grip
- 11  Water tube and bite valve (shown cut)
- 12  Equipment plate and spacers
- 13  Flow meter and bracket
- 14  Battery, blower, control modules
- 15  Case, drilled, with wall fittings and filter
- 16  Lid with monitor bracket and monitor
- 17  Reservoir stand and bottle



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1781/voidscope

Figure 1. Every component pulled apart and numbered in build order. The rod sections lie side by side as they are stored; the cable and tubes are drawn cut short.

The prototype is a 4.4 m camera probe and its surface unit. The probe is four aluminium tube sections that click together with snap buttons, a turned aluminium head holding a sealed camera, a small speaker and a bite valve, and a turned tail cap where the air goes in. The camera cable and a guide tube for the water stay threaded through the sections. The surface unit is a bought hard case holding a battery, a blower, a flow meter and small control modules, with a monitor in its lid; a water bottle hangs on a light stand beside it. Figure 1 shows the 17 components in the order you make or fit them. Nine are made: the rod sections, couplers, collar, nose and tail cap (tube cutting, drilling and lathe work), the equipment plate and two small brackets (sheet and angle), and the drilled case. The rest are bought and fitted. The parts cost about USD 1,578 from the bill of materials.

Safety: VoidScope is rescue equipment for trained teams; this prototype is not certified and it is not a medical device. The battery packs are 12.8 V lithium iron phosphate: keep the battery lead unplugged and the fuse out until section 6 says otherwise, and charge packs only on their own charger, on a non-combustible surface. Turning, drilling and cutting tube throw chips and leave sharp edges: wear safety glasses, keep hair and sleeves clear of the lathe chuck, and deburr everything. The water line must never be connected to a pump or a pressurised container, and the capillary is never removed. Two-part epoxy and threadlocker need gloves and ventilation.

■ 2. What changed to make it buildable

The concept showed what the probe does; it did not say how the sections join, how the lines cross the joints or how the head fits everything inside 48 mm. Each change keeps what VoidScope does, and all of them are recorded in decision record VDS-DDR-002.

Table 1. Changes from the concept.

Component	The concept had	The buildable design has	Why
Rod and joints	A rod about 50 mm across in about 1 m sections; joints not defined	38.1 mm tube in four 1,050 mm sections, a 34.93 mm coupler bonded and pinned in each, locked by a snap button (Figure 3)	The two stock tube sizes slide together by hand, and the thinner rod leaves room for the head
Lines	Connectors at each joint	Cable and guide tube stay threaded through every section, like the cord in a tent pole	No connectors inside the rod; setup is four clicks
Air	A separate air line	The rod's own bore carries the air	Far less pressure loss; the joints need no seals
Water	A line inside the rod	A guide tube the single-use water tube pushes through (Figure 9)	The water set changes in about 4 minutes, no tools
Head	One sealed piece	A turned collar and nose joined by three screws; camera set 6.5 mm below the centre; a 20 mm speaker (Figures 5 to 8)	Each part turns on a small lathe; the bite valve and speaker fit inside 48 mm
Display	A handheld screen	A monitor in the lid of the surface unit (Figure 16)	One battery and one cable; the crew's hands stay on the rod
Surface unit	Not laid out	An equipment plate on spacers, an upright flow meter, sealed wall fittings (Figure 13)	The case stays sealed and the lid still closes
Details found by checking the model	A 25 mm tail cap bore, a water groove 0.5 mm too deep, a short valve pocket	A 30 mm bore with a 44 mm flange, the groove floor 20 mm from the centre, a 50 mm pocket	Each part now clears its neighbours

3. Making the components

Make and check each component before the assembly step that needs it. Sizes are in millimetres. "Top" is the side the bite valve pocket is on; the "button line" is a straight line scribed along every section and coupler, on the right side as seen from behind the tail cap, so all the buttons line up. Workshop tolerance is 0.2 mm on turned diameters and 0.5 mm elsewhere unless a step says otherwise; drawings do not carry tolerances before TRL 4.

3.1 Rod sections (make 4)

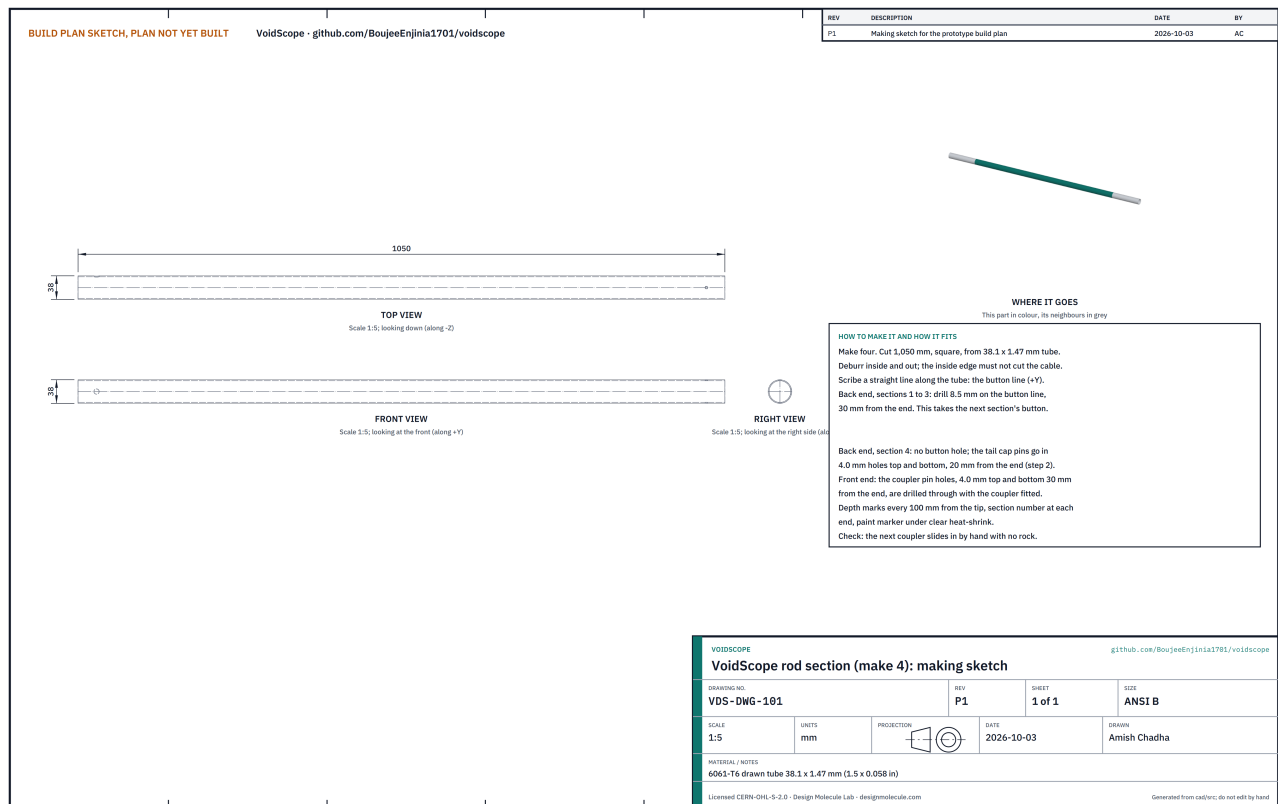


Figure 2. Rod section making sketch (VDS-DWG-101).

What it is and what it is made from. The four straight lengths of the probe. Each is 6061-T6 drawn aluminium tube, 38.1 mm (1.5 in) outside with a 1.47 mm (0.058 in) wall, 1,050 mm long.

How to make it.

1. Cut four lengths of 1,050 mm, square, with a fine-tooth saw or a tube cutter. Number them 1 to 4; section 1 goes next to the head.
2. Deburr both ends inside and out. The inside edge must be smooth, because the cable slides past it.
3. Scribe the button line along each tube with a straight edge.
4. Sections 1, 2 and 3: on the button line, 30 mm from the back end, drill an 8.5 mm hole (start with 3 mm, then step up). The next section's snap button comes through this hole.
5. Section 4: no button hole at the back. The tail cap pins go through it in step 2.
6. Mark depth bands every 100 mm, counting from the tip of the assembled probe, and the section number at each end, with a paint marker. Cover each section with clear heat-shrink.

How it fits the parts next to it. The front end of each section takes a coupler, bonded in for 60 mm (Figure 3). The back end slides over the coupler of the section behind it until the two tube ends butt; the snap button pops through the 8.5 mm hole. Section 1's front coupler goes into the collar instead.

Check before moving on. A coupler slides into each back end by hand without rocking, and all four sections are 1,050 mm within 0.5 mm.

3.2 Couplers with snap buttons (make 4)

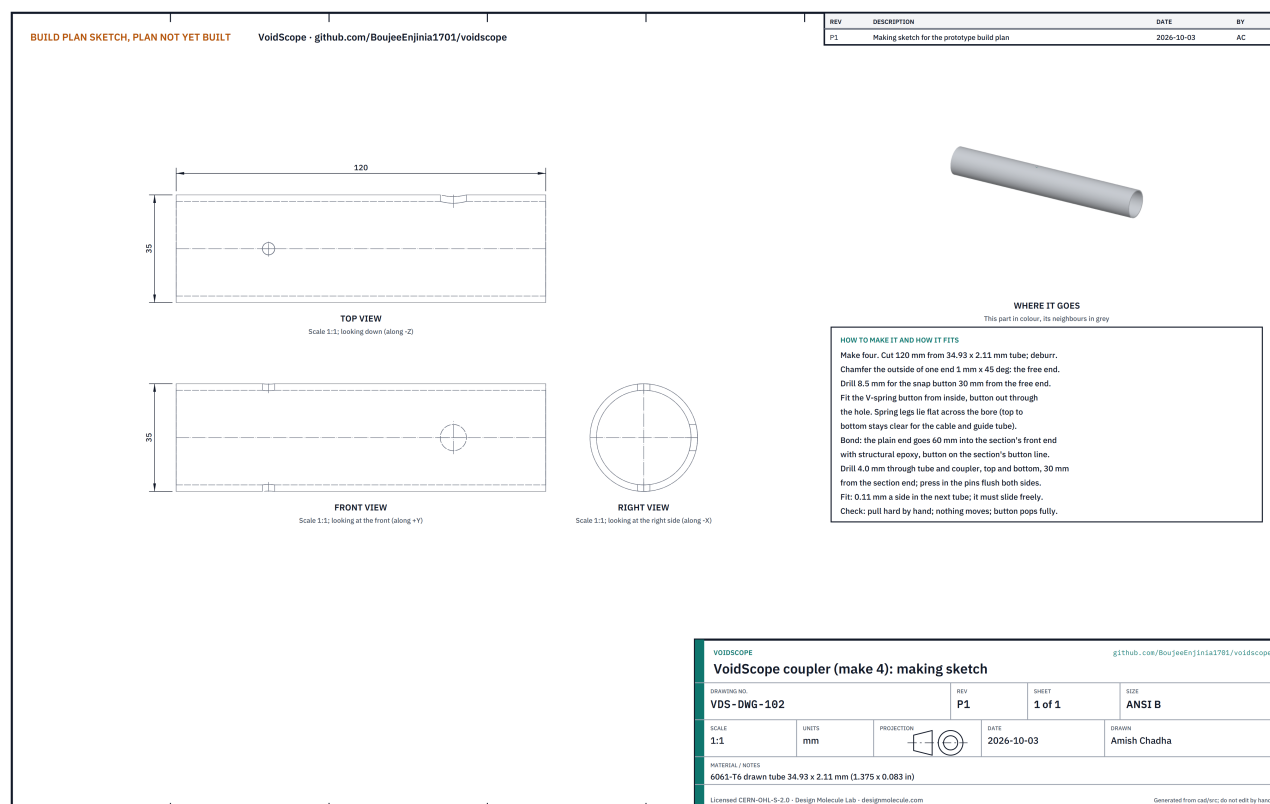


Figure 3a. Coupler making sketch (VDS-DWG-102).

What it is and what it is made from. The short inner tube that joins two sections. 6061-T6 tube, 34.93 mm (1.375 in) outside with a 2.11 mm (0.083 in) wall, 120 mm long; a stainless V-spring snap button with an 8 mm button; two 4 mm aluminium pins.

How to make it.

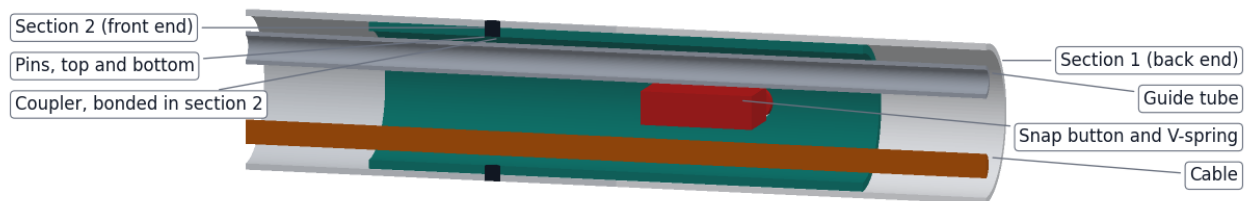
1. Cut four 120 mm lengths; deburr. Chamfer the outside of one end 1 mm at 45°: this is the free end.
2. Scribe a button line along each. On it, 30 mm from the free end, drill an 8.5 mm hole.
3. Squeeze the V-spring and slide it in from the plain end, button first, until the button snaps out through the hole. The spring strip lies across the bore from the button side to the opposite side; the spaces above and below it are where the cable and guide tube pass.
4. Bond: clean the plain end and the inside of the section's front end with solvent, coat both with structural epoxy, and push the coupler 60 mm into the section with the two button lines in line. Wipe off the squeeze-out inside.

- Before the epoxy sets, drill 4.0 mm through tube and coupler at the top and the bottom, 30 mm from the section's front end, and press in the two pins flush both sides. Leave to cure for 24 hours.

How it fits the parts next to it.

Joint 1: two rod sections joined by the coupler and snap button

Cut on the vertical centre plane, seen from the side. Tube ends butt; the button locks; cable and guide pass the spring



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/voidscope

Figure 3. The coupler is bonded and pinned into one section and slides into the next; the tube ends butt and the button locks. The cable and guide tube pass above and below the spring.

The coupler's outside slides inside the next tube with about 0.11 mm a side. Push loads go through the butted tube ends; pull and twist go through the button. The spring sits 3.0 mm from the cable and 2.0 mm from the guide tube.

Check before moving on. Each joint clicks together and comes apart with one press of the button; a hard pull by hand moves nothing.

3.3 Collar

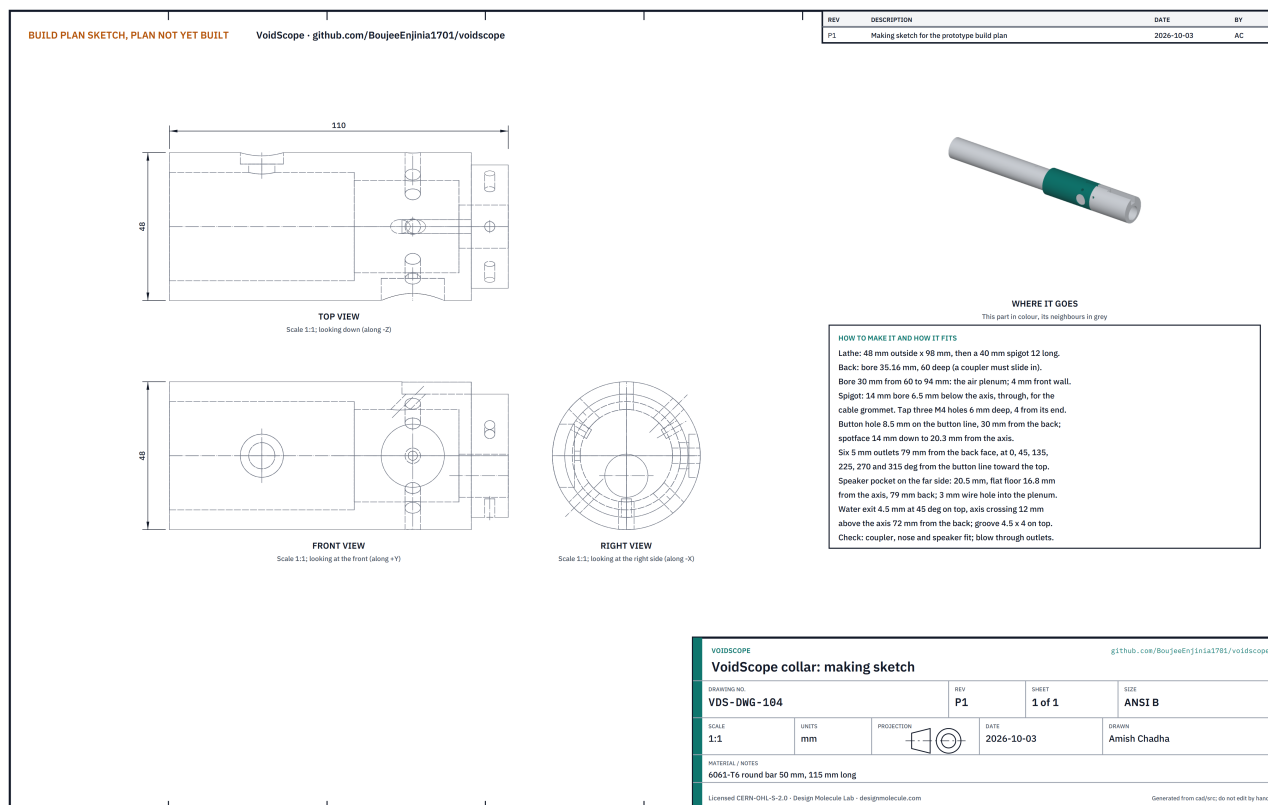


Figure 4. Collar making sketch (VDS-DWG-104).

What it is and what it is made from. The back half of the head: it takes the first rod section, turns the air out through six holes, holds the speaker, lets the water tube out and carries the nose. 6061-T6 round bar, 50 mm, 115 mm long.

How to make it.

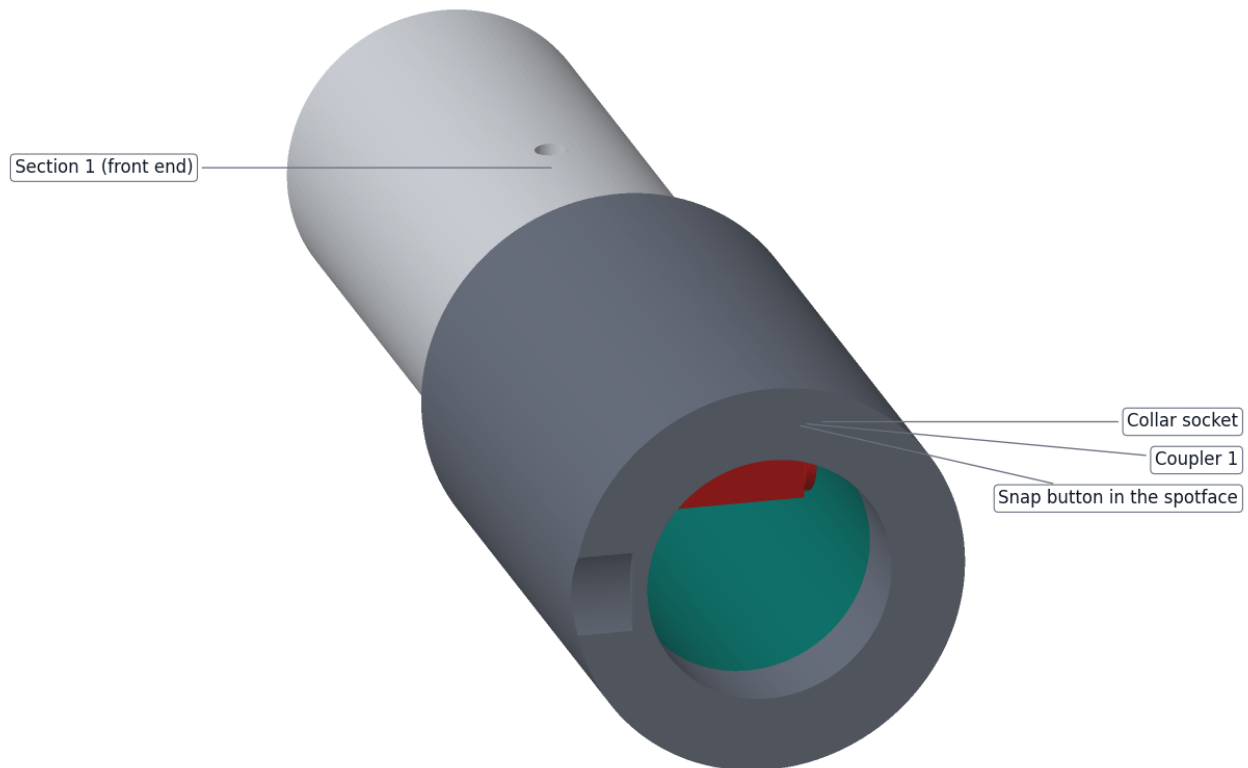
1. On the lathe, turn to 48 mm outside for 98 mm and a 40 mm spigot 12 mm long at the front; part off.
2. From the back face, bore 35.16 mm, 60 mm deep, so a coupler slides in. Then bore 30 mm on to 94 mm from the back face: the air chamber. Leave a 4 mm front wall.
3. Through the spigot, bore 14 mm on a centre 6.5 mm below the axis (offset in a four-jaw chuck, or drill and ream on a drill press with the collar in a vee block). This takes the cable grommet.
4. Tap three M4 holes 6 mm deep into the spigot, 4 mm from its front end, at the positions on the sketch.
5. On the button line, 30 mm from the back face, drill 8.5 mm, then spotface 14 mm across until the floor is 20.3 mm from the axis, so a fingertip reaches the button.
6. Drill six 5 mm air holes into the chamber, 79 mm from the back face: on the button line, at 45° and 135° above it, and at 225°, 270° and 315° (bottom and the lower side).
7. On the side opposite the button line, at 79 mm from the back face, mill a 20.5 mm flat-bottomed pocket down to 16.8 mm from the axis for the speaker; drill a 3 mm wire hole from its floor into the chamber.

8. On top, drill the 4.5 mm water exit at 45° through into the chamber, its axis crossing 12 mm above the axis at 72 mm from the back face (use a 45° vee block on the drill press). Mill a groove 4.5 mm wide on top from the exit to the front face, its floor 20 mm above the axis.
9. Deburr every hole inside and out; the water tube and the air must meet no sharp edge.

How it fits the parts next to it.

Joint 2: section 1 into the collar socket

Seen from the button side. The button stands just proud in a 14 mm spotface so a fingertip can press it



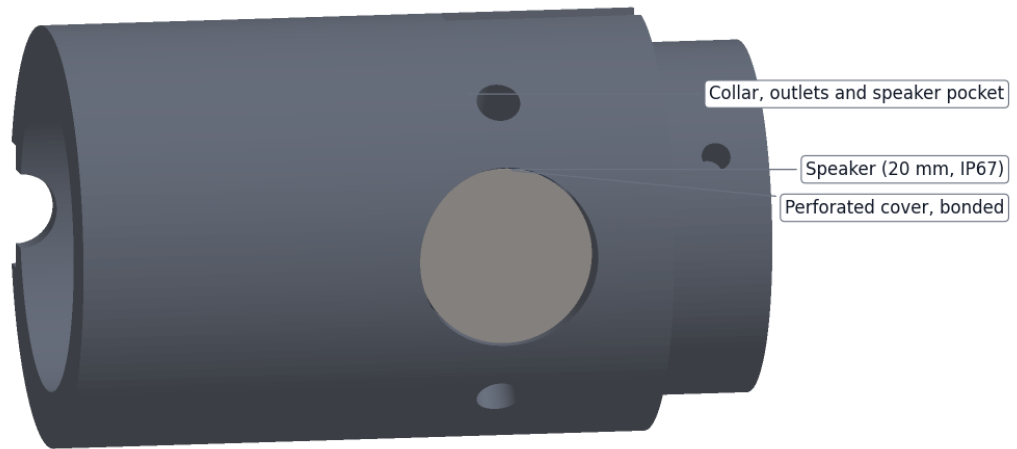
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/voidscope

Figure 5. Section 1's coupler slides into the collar socket until the tube butts the collar; the button stands just proud in its spotface.

Joint 5: speaker in its pocket on the collar

Seen from the speaker side. Flat pocket floor; cover flush below the 48 mm outline; outlets at 45 deg either side



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/voidscope

Figure 6. The 20 mm speaker sits on the flat floor of its pocket under a bonded perforated cover, flush inside the 48 mm outline; air holes lie 45° either side.

The socket takes section 1 exactly like a rod joint. The 40 mm spigot fits the nose's 40.2 mm counterbore. The speaker is bedded in neutral-cure silicone and its cover bonded on top.

Check before moving on. A coupler slides into the socket and the button pops into the spotface. Air blown into the socket comes out of all six holes.

3.4 Nose

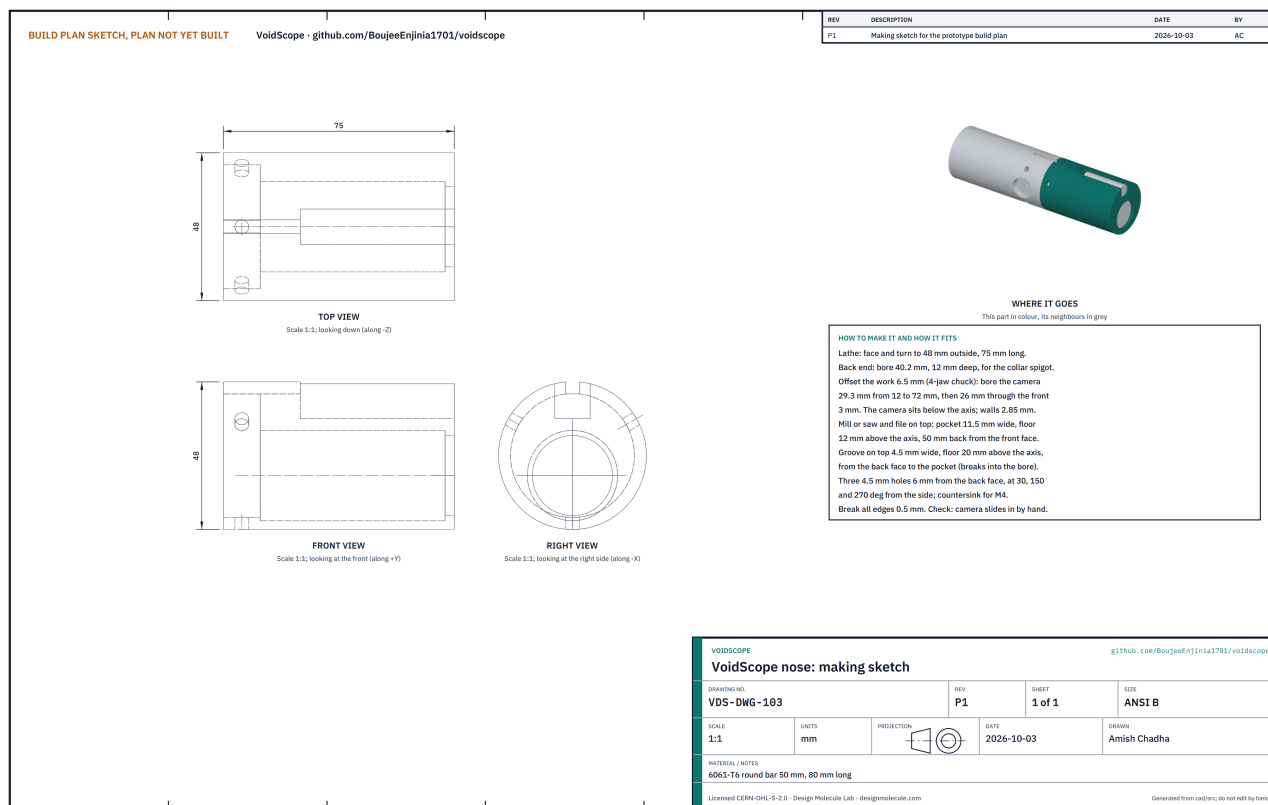


Figure 7. Nose making sketch (VDS-DWG-103).

What it is and what it is made from. The front of the head: it holds the camera behind a protecting lip and parks the bite valve on top. 6061-T6 round bar, 50 mm, 80 mm long.

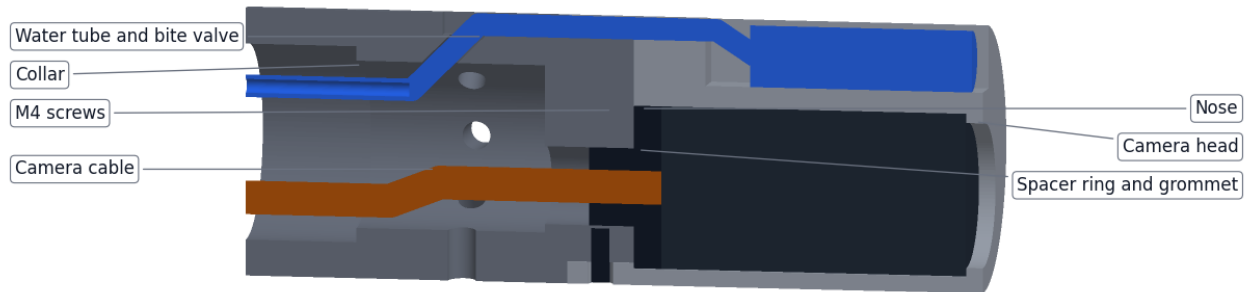
How to make it.

1. Turn to 48 mm outside and 75 mm long. From the back face, bore 40.2 mm, 12 mm deep, to fit the collar spigot.
2. Set the work 6.5 mm off centre in a four-jaw chuck, the offset away from the top. Bore 29.3 mm from 12 mm to 72 mm from the back face, then 26 mm through the last 3 mm: the lip the camera rests against. The thinnest wall is 2.85 mm, at the bottom.
3. On top, cut the bite valve pocket: 11.5 mm wide, its floor 12 mm above the axis, 50 mm long, open at the front. A mill is best; a saw and file will do.
4. Continue a groove 4.5 mm wide on top, floor 20 mm above the axis, from the back face to the pocket. Over the counterbore it breaks through; that is intended.
5. Drill three 4.5 mm holes 6 mm from the back face at the same angles as the tapped holes in the spigot, and countersink them for M4 screws.
6. Break every edge by 0.5 mm, especially at the front lip and the pocket.

How it fits the parts next to it.

Joint 3: nose on the collar spigot, camera clamped against the lip

Cut on the vertical centre plane. The spacer ring pushes the camera onto the 26 mm lip; three M4 screws hold the nose



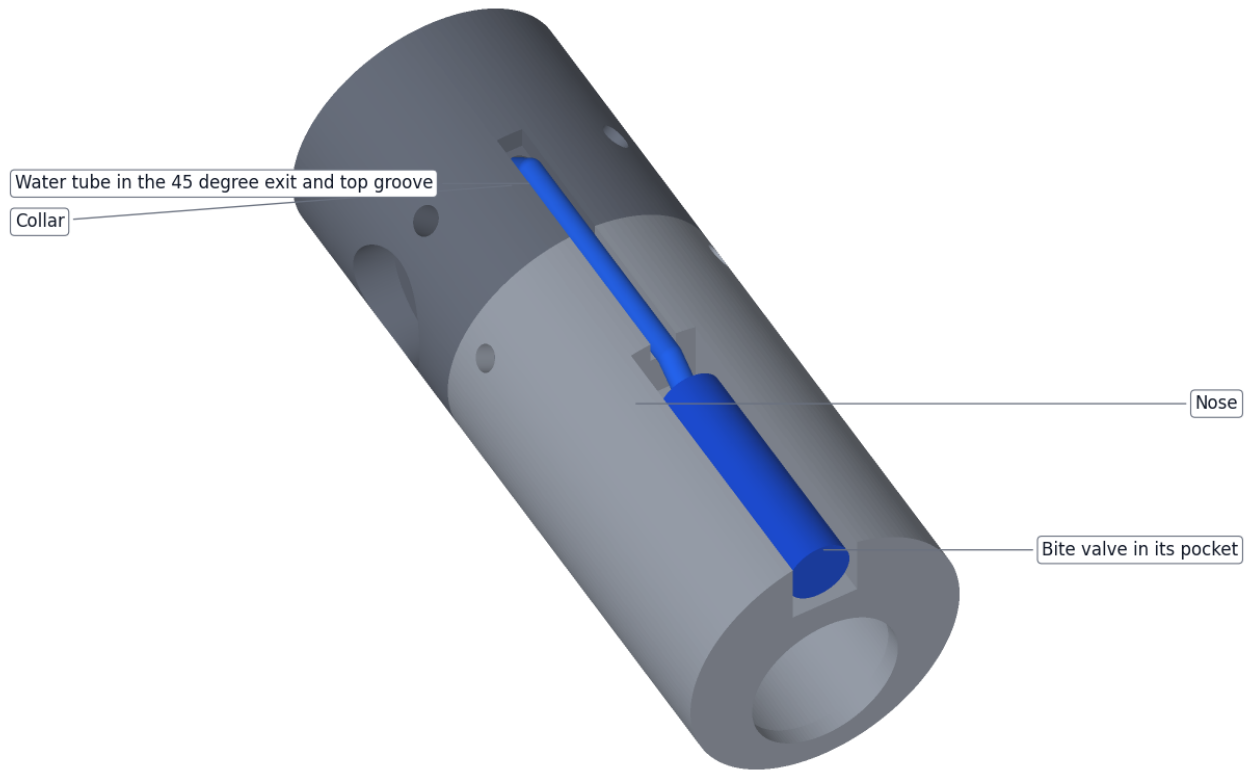
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/voidscope

Figure 8. The camera slides into the nose from behind until its face rests on the lip; a rubber spacer ring between the camera and the spigot clamps it; three M4 screws hold the nose on the spigot.

Joint 4: water tube out of the collar, along the groove, into the bite valve pocket

Seen from above and in front. The tube lies flush in a 4.5 mm groove; the valve presses into an 11.5 mm pocket



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/voidscope

Figure 9. The water tube leaves the collar through the 45° exit, lies flush in the top groove and ends in the bite valve, pressed into its pocket on the nose.

Check before moving on. The camera slides in by hand and its face sits 3 mm behind the front of the nose. The nose slides onto the spigot and the three screw holes line up. A bite valve presses into the pocket and stays there when the nose is shaken.

3.5 Tail cap

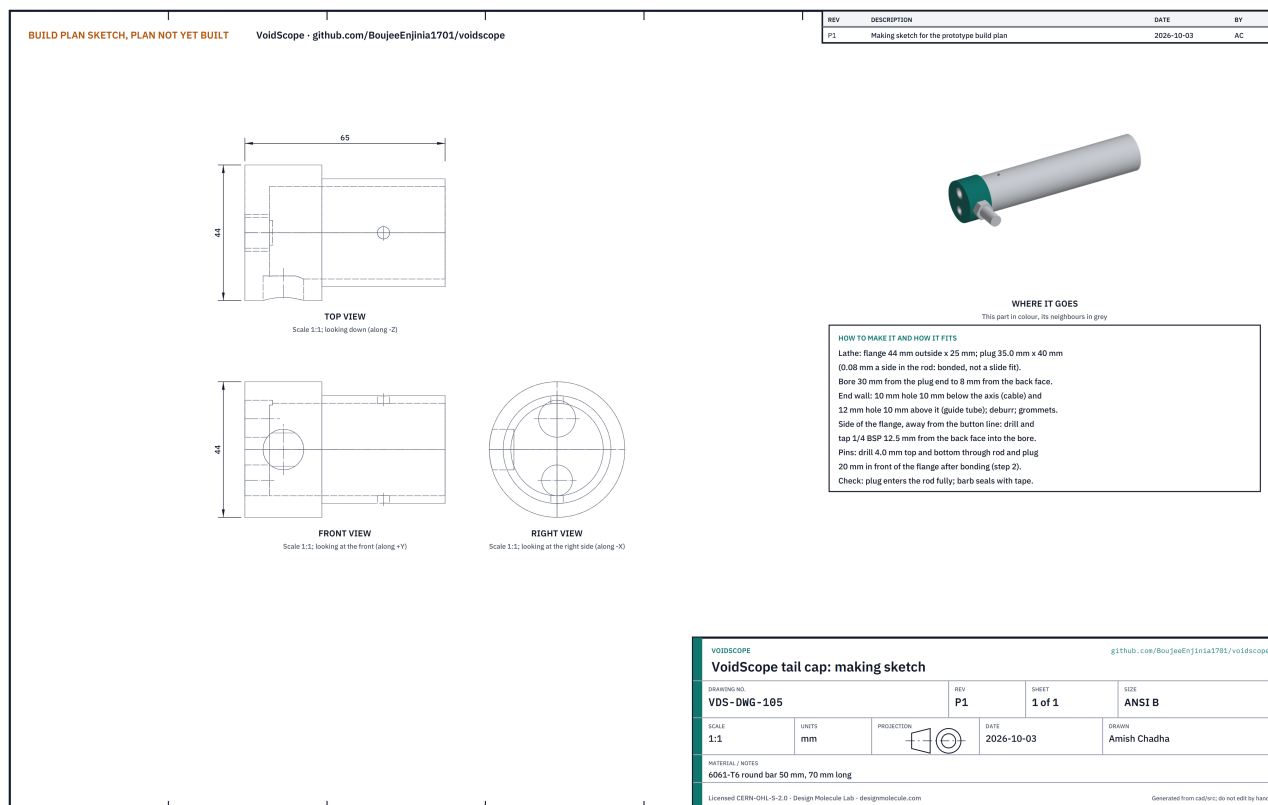


Figure 10. Tail cap making sketch (VDS-DWG-105).

What it is and what it is made from. The plug in the back end of section 4: the air goes in here, and the cable and guide tube come out through two grommets. 6061-T6 round bar, 50 mm, 70 mm long; a brass 1/4 BSP to 12 mm hose barb; two split rubber grommets; two 4 mm pins.

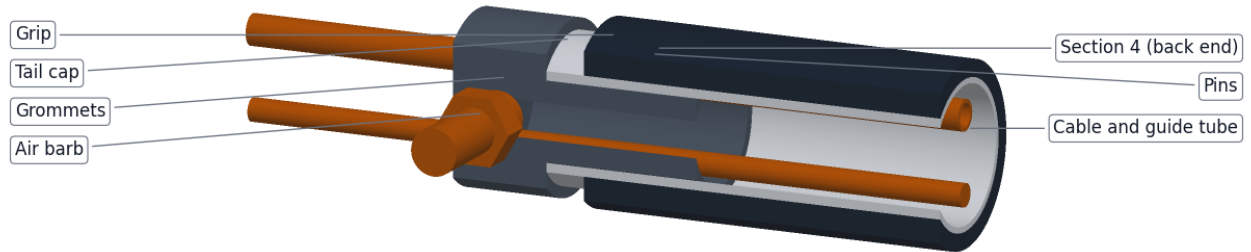
How to make it.

1. Turn a 44 mm flange 25 mm long and a 35.0 mm plug 40 mm long. The plug is 0.08 mm a side smaller than the tube's bore: a bonded fit, not a slide fit.
2. Bore 30 mm from the plug end to 8 mm from the back face, leaving an 8 mm end wall.
3. In the end wall, drill a 10 mm hole 10 mm below the axis for the cable grommet and a 12 mm hole 10 mm above it for the guide tube grommet. Deburr. Split each grommet on one side so the lines can be laid in.
4. In the side of the flange opposite the button line, 12.5 mm from the back face, drill and tap 1/4 BSP into the bore. Fit the barb with thread-seal tape.

How it fits the parts next to it.

Joint 6: tail cap bonded and pinned in section 4

Cut on the vertical centre plane. Air enters by the barb; cable and guide slide through split grommets



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/voidscope

Figure 11. The plug is bonded into section 4 against the flange and pinned top and bottom; air comes in by the barb; the cable and guide tube slide through the grommets.

Check before moving on. The plug goes fully into section 4 dry. Blowing into the barb with a thumb over each grommet, air comes out of the plug end only.

3.6 Equipment plate

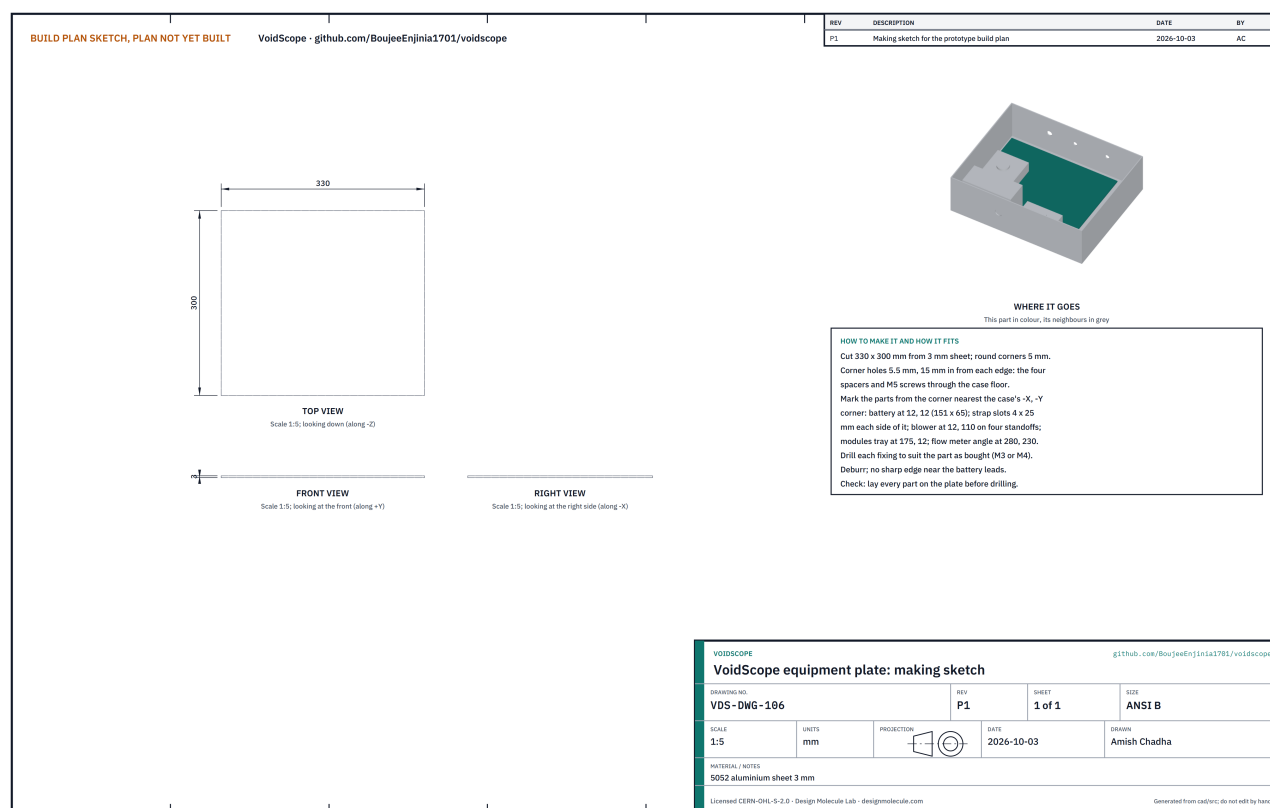


Figure 12. Equipment plate making sketch (VDS-DWG-106).

What it is and what it is made from. The flat plate in the bottom of the case that every surface unit part is fixed to. 5052 aluminium sheet, 3 mm, 330 x 300 mm.

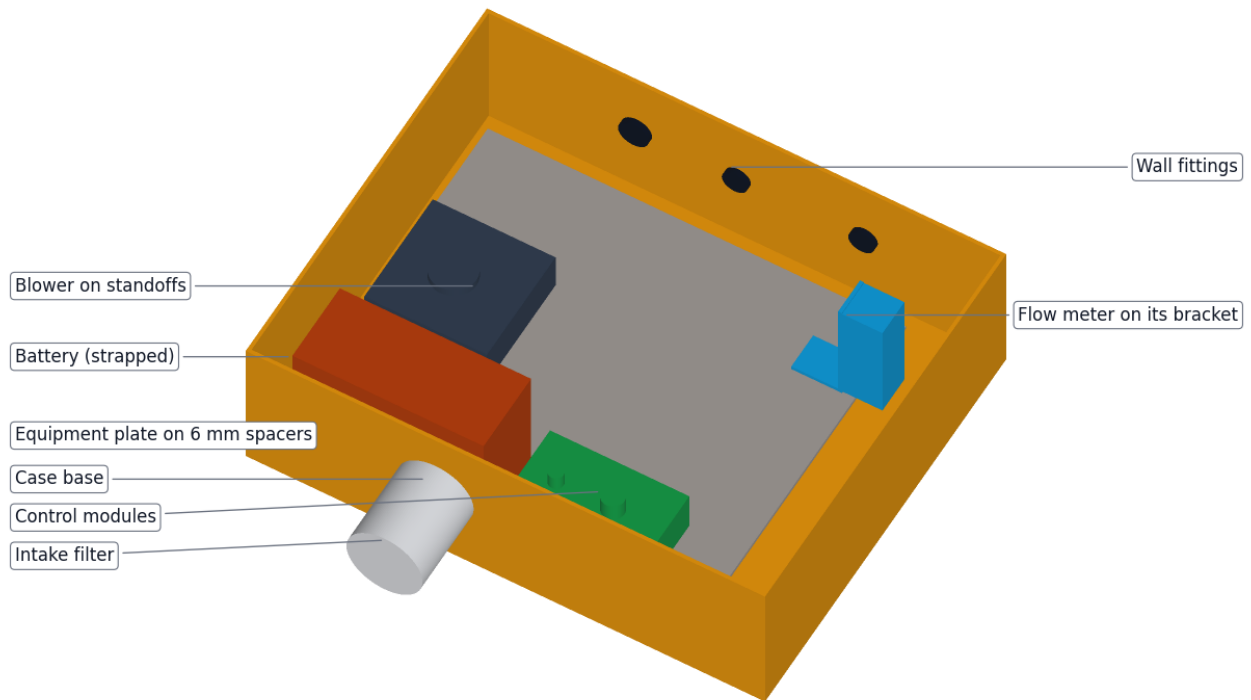
How to make it.

1. Cut the blank; round the corners to 5 mm; deburr.
2. Drill four 5.5 mm holes, 15 mm in from each edge at the corners.
3. Lay out the parts from the corner that will sit nearest the case's back left corner: the battery 12 mm in from both edges, with two pairs of 4 x 25 mm strap slots beside it; the blower 12 mm from the left edge and 110 mm from the front, on four standoffs; the control modules tray 175 mm from the left edge; the flow meter bracket 280 mm from the left edge and 230 mm from the front.
4. Drill each fixing hole to suit the part as bought (M3 or M4).

How it fits the parts next to it.

Joint 7: inside the surface unit base

Seen from above, lid not shown. Everything stands on the plate; the tallest part is 19 mm below the rim



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/voidscope

Figure 13. The plate stands on four 6 mm spacers, screwed through the floor; everything else stands on the plate. The tallest part is 19 mm below the rim.

Check before moving on. Every part sits on the plate without overlapping another, and the plate drops into the case with 6 mm clear of every wall.

3.7 Flow meter bracket

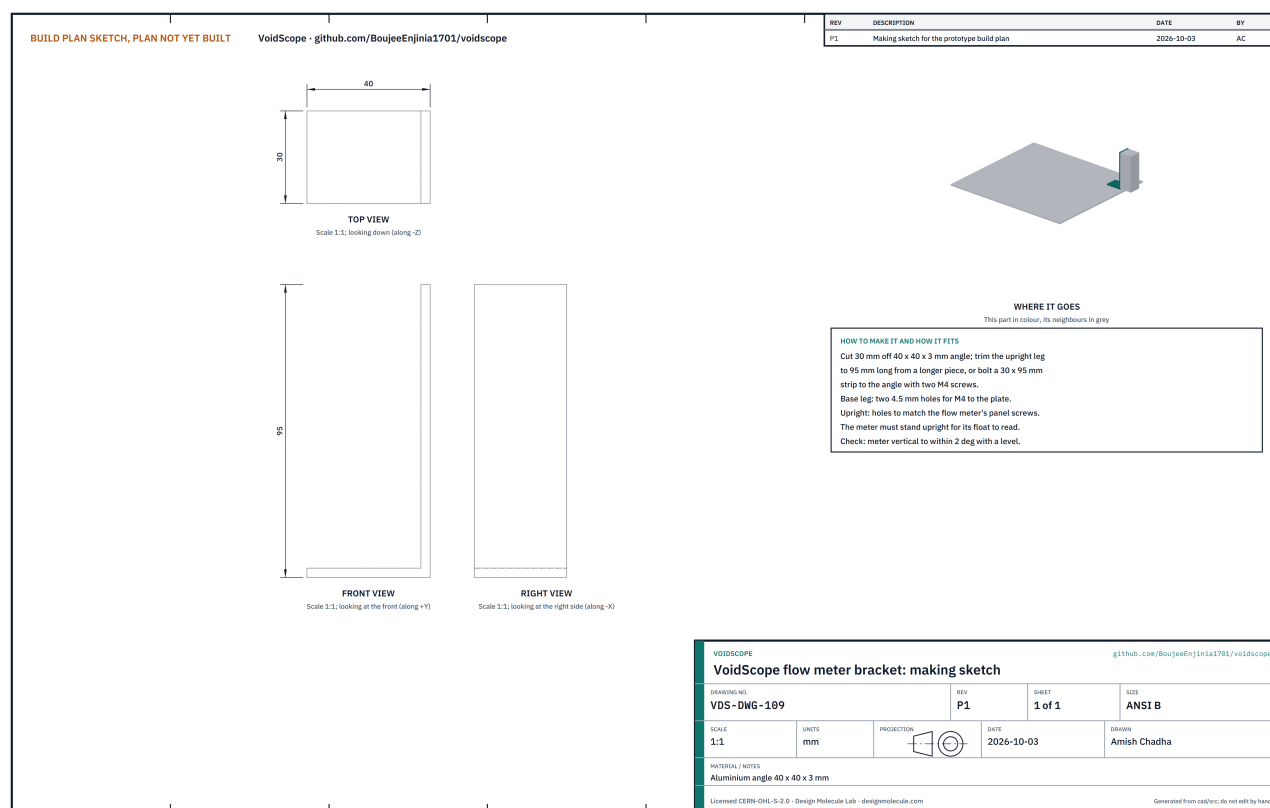


Figure 14. Flow meter bracket making sketch (VDS-DWG-109).

What it is and what it is made from. A short aluminium angle that holds the flow meter upright. Aluminium angle 40 x 40 x 3 mm.

How to make it. Cut 30 mm of angle and extend the upright leg to 95 mm with a bolted strip if the angle is too short. Drill two 4.5 mm holes in the base leg and holes in the upright to match the flow meter's panel screws.

How it fits the parts next to it. The base leg bolts to the plate with two M4 screws; the meter screws to the upright. The meter must stand upright for its float to read.

Check before moving on. With the plate on a level bench, the meter is vertical within 2° on a spirit level.

3.8 Monitor bracket

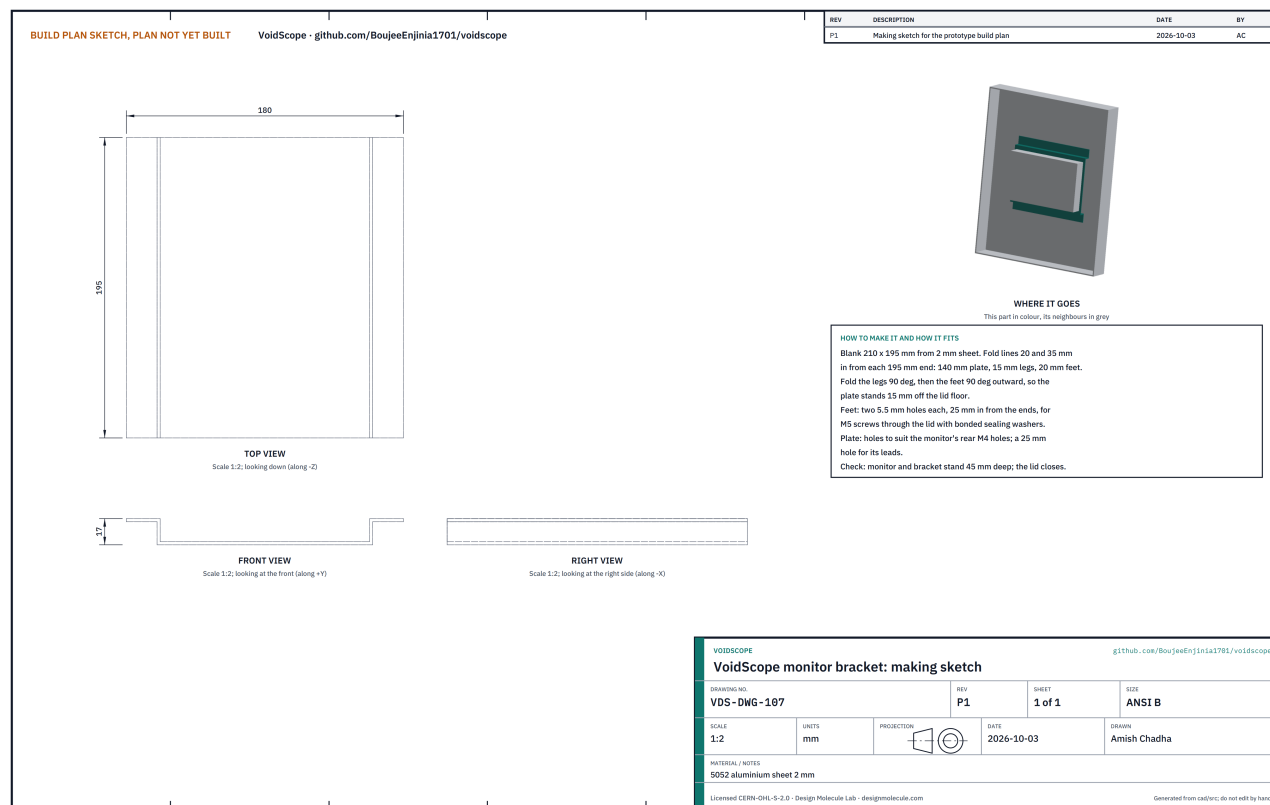


Figure 15. Monitor bracket making sketch (VDS-DWG-107), drawn as it sits with the lid closed.

What it is and what it is made from. A folded plate that holds the monitor 15 mm off the lid floor. 5052 aluminium sheet, 2 mm.

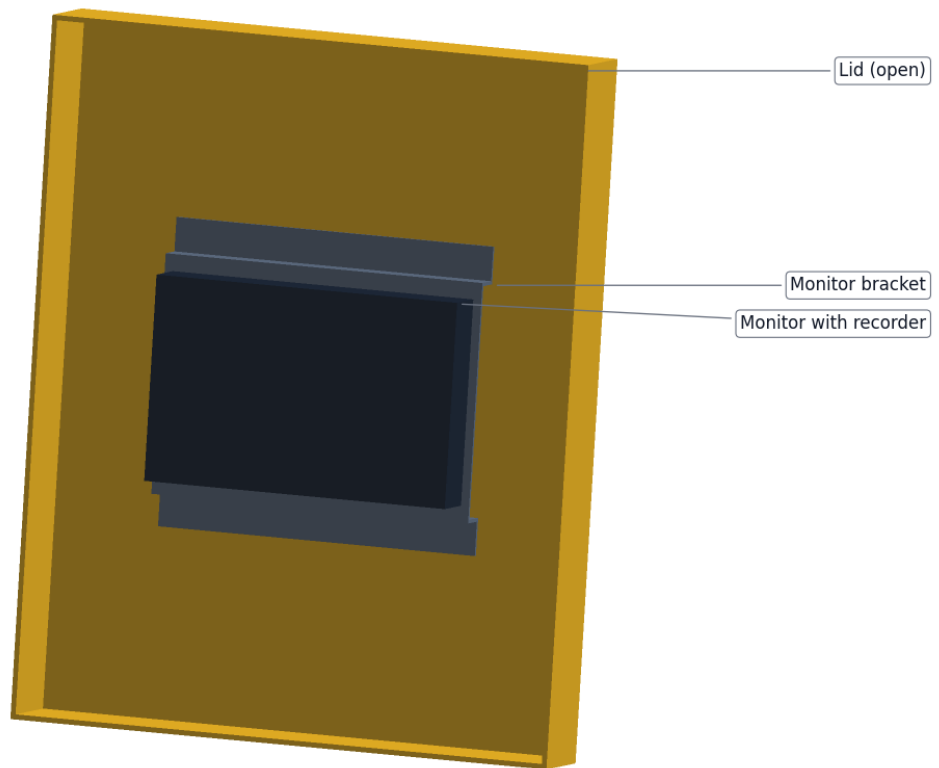
How to make it.

1. Cut a 210 x 195 mm blank. Mark fold lines 20 mm and 35 mm in from each 195 mm end.
2. Fold the 35 mm lines 90° up (the legs) and the 20 mm lines 90° out (the feet).
3. Drill two 5.5 mm holes in each foot, 25 mm in from its ends; drill the plate to match the monitor's rear M4 holes and a 25 mm hole for its leads.

How it fits the parts next to it.

Joint 8: monitor on its bracket in the lid

Seen from the operator's side. Bracket feet bolted through the lid with sealing washers



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/voidscope

Figure 16. The bracket's feet bolt through the lid; the monitor screws to the bracket's plate and faces the operator when the lid is open.

Check before moving on. Bracket and monitor together stand 45 mm deep or less, so the lid still closes.

3.9 Surface unit case, drilled

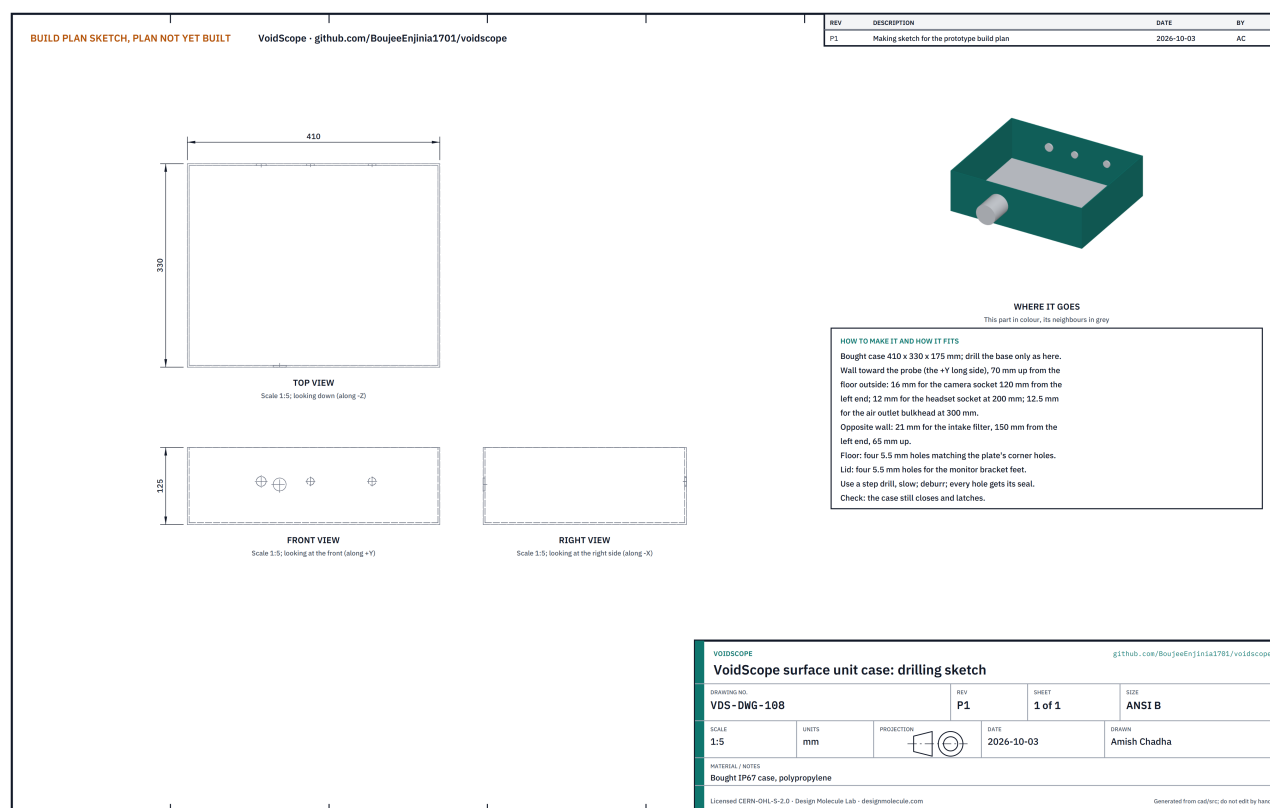


Figure 17. Case drilling sketch (VDS-DWG-108).

What it is and what it is made from. A bought IP67 hard case, about 410 x 330 x 175 mm outside, drilled for the plate, the wall fittings and the filter, and its lid drilled for the monitor bracket.

How to make it.

1. On the long wall that will face the probe, 70 mm up from the floor outside: drill 16 mm for the camera socket 120 mm from the left end, 12 mm for the headset socket at 200 mm and 12.5 mm for the air outlet at 300 mm.
2. On the opposite wall, drill 21 mm for the intake filter, 150 mm from the left end and 65 mm up.
3. In the floor, drill four 5.5 mm holes to match the plate's corner holes (use the plate as a template). In the lid, drill four 5.5 mm holes to match the bracket feet.
4. Use a step drill at low speed so the plastic does not crack; deburr every hole.

How it fits the parts next to it. Every fitting goes through its hole with its seal on the outside and its nut inside; every screw through the floor or lid has a bonded sealing washer, so the case stays sealed.

Check before moving on. The case closes and latches with nothing fitted.

3.10 Bought components

- **Camera head:** sealed pipe-inspection head, 29 mm across, about 55 mm long, IP68, 1080p (AHD), about 90° view, 12 dimmable LEDs, 12 V. Trim its pigtail to 80 mm.

- **Speaker:** 20 mm IP67 micro speaker, 8 Ω , 0.5 W, about 4 mm thick; a 20 mm disc of perforated stainless sheet as its cover.
- **Cable:** hybrid camera cable, 7 m, with one mini-coax for video and four small cores, 6 mm across at most, with a six-pin aviation plug fitted at the surface end.
- **Guide tube:** PTFE, 8 mm outside, 6 mm bore, 6.5 m.
- **Water set (single use):** food-grade polyurethane tube 4 mm by 2.5 mm, 8 m; silicone bite valve about 11 mm across; 600 mm of 0.5 mm bore food-grade PTFE capillary with two barbed reducers; roller clamp; three-way stopcock; 100 micron strainer; 60 mL syringe. Keep it sealed in its bag until use.
- **Grip:** rubber or foam handlebar grip tube, 38 mm bore, 300 mm.
- **Battery packs (2):** 12.8 V 6 Ah LiFePO₄ with built-in protection and an XT60 lead, about 151 x 65 x 94 mm, with a hook-and-loop strap.
- **Blower:** 12 V centrifugal, about 97 x 94 x 33 mm, shut-off pressure 1.0 kPa or less, PWM speed input.
- **Intake filter, flow meter, control modules, wall fittings, headset:** as in the bill of materials, lines 19 to 23.
- **Monitor:** 7 in, AHD input, recorder, 1000 nit or brighter, 12 V, rear M4 holes, 28 mm deep or less.
- **Reservoir:** 1 L graduated food-grade bottle with a bail and a bottom spigot; light stand 0.8 to 2.0 m with a hook arm; mark the stand every 0.1 m.
- **Air hose and sleeve, charger, rod carry case, cleaning kit:** as in the bill of materials, lines 24 to 27.

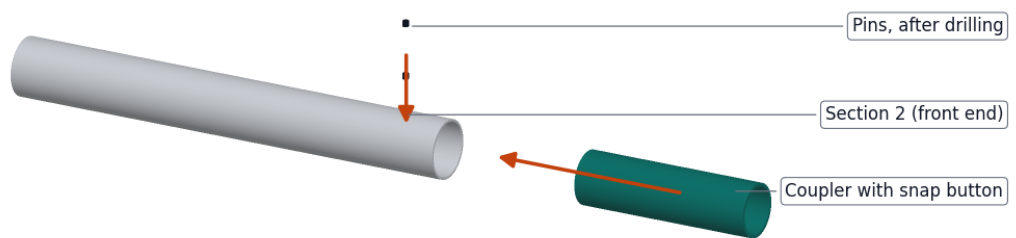
■ 4. Putting it together

In each picture the parts already fitted are grey and the part being fitted is in colour, with an arrow showing the way it goes in. Long parts are drawn cut short where the whole length would make the picture unreadable.

Step 1: couplers into the sections

Step 1: coupler into each section (do this four times)

Epoxy on the plain end; push it 60 mm in, button on the button line; drill and pin top and bottom; cure 24 h



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

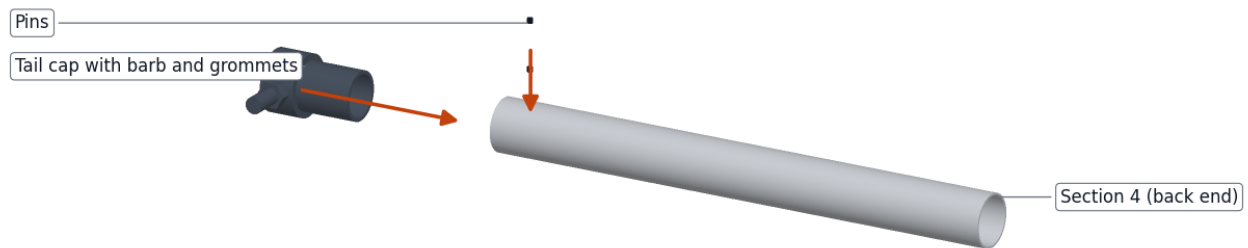
github.com/BoujeeEnjinia1701/voidscope

Done for each of the four sections in section 3.2: epoxy, push in 60 mm with the button lines in line, drill and pin top and bottom, cure for 24 hours. Hold point: do not join any section until the epoxy has cured.

Step 2: tail cap into section 4

Step 2: tail cap into section 4

Epoxy on the plug; push it home against the flange, barb away from the button line; drill and pin



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

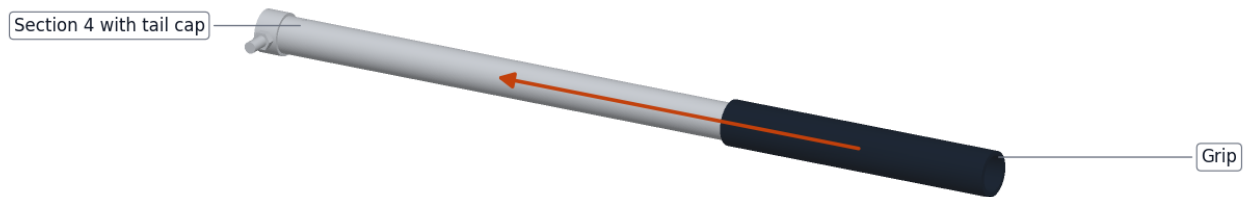
github.com/BoujeeEnjinia1701/voidscope

Epoxy on the plug and in the back end of section 4; push home against the flange with the barb opposite the button line; drill 4.0 mm through tube and plug top and bottom, 20 mm in front of the flange, and press in the pins.

Step 3: grip onto section 4

Step 3: grip onto section 4

Wet the inside with soapy water; slide it on from the front end until it is 10 mm from the flange



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

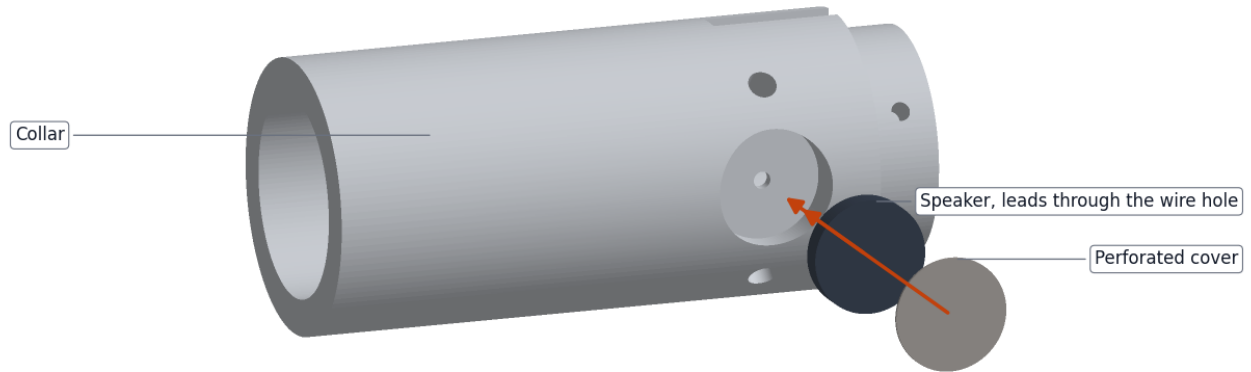
github.com/BoujeeEnjinia1701/voidscope

Wet the inside of the grip with soapy water and slide it on from the front end of section 4 until it stops 10 mm in front of the flange. Let it dry.

Step 4: speaker into the collar

Step 4: speaker into the collar

Neutral-cure silicone on the pocket floor; leads into the plenum; cover bonded flush



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

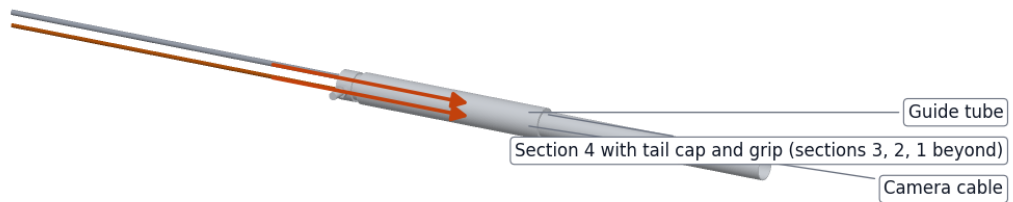
github.com/BoujeeEnjinia1701/voidscope

Feed the speaker leads through the 3 mm wire hole into the chamber. Bed the speaker on neutral-cure silicone, then bond the perforated cover over it, flush.

Step 5: thread the cable and guide tube through every section

Step 5: thread the cable and guide tube through every section

From the tail cap end, through both grommets and on through sections 4, 3, 2 and 1; 1 m of each left out at the front. Lines drawn cut short



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

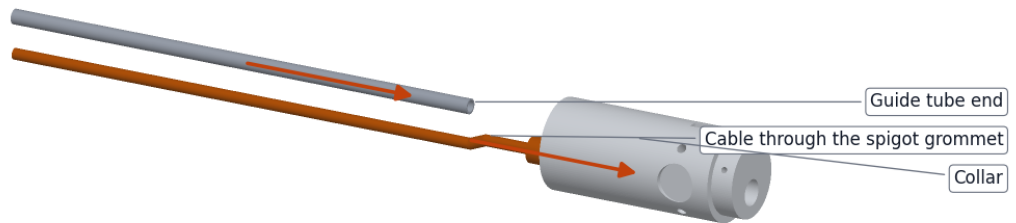
github.com/BoujeeEnjinia1701/voidscope

Lay the sections in a line, tail cap at the back. From the tail cap end, feed the cable and the guide tube through their grommets and through sections 4, 3, 2 and 1 in turn, staying in the spaces above and below each spring. Leave 1 m of each out at the front and about 2 m behind the tail cap.

Step 6: cable and guide tube into the collar

Step 6: cable and guide into the collar

Guide 6 mm into the plenum; cable through the grommet; solder to the camera pigtail and speaker leads, heat-shrink



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

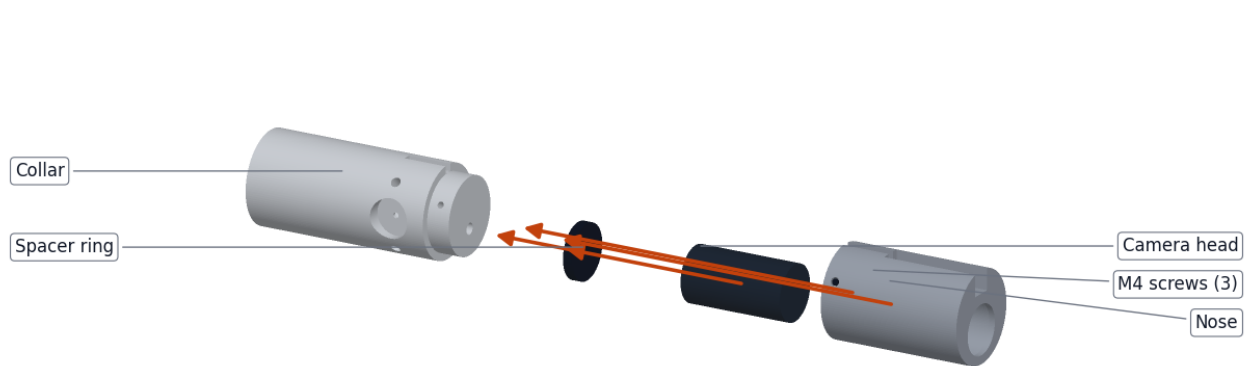
github.com/BoujeeEnjinia1701/voidscope

Push the guide tube 6 mm into the collar's air chamber. Pass the cable through the spigot grommet; solder its video and power cores to the camera pigtail and its speaker pair to the speaker leads, and cover every joint with heat-shrink. Seal the grommet with a little silicone.

Step 7: camera and nose onto the collar

Step 7: camera and nose onto the collar

Camera into the nose from behind, LEDs against the lip; ring next; nose over the spigot; three M4 screws, threadlocker



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

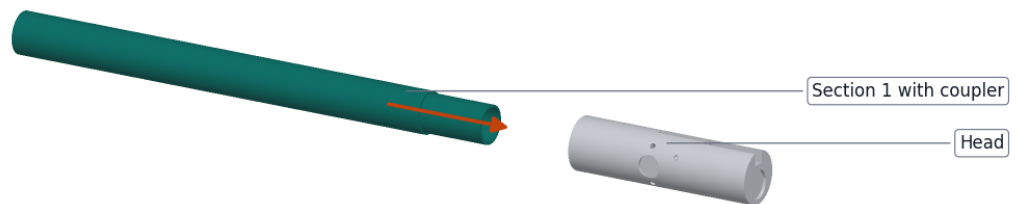
github.com/BoujeeEnjinia1701/voidscope

Slide the camera into the nose from behind until its face rests on the lip; fit the spacer ring behind it; slide the nose onto the spigot so the ring clamps the camera. Three M4 countersunk screws with threadlocker.

Step 8: section 1 into the collar

Step 8: section 1 into the collar

Press the button, slide the coupler home until the tube butts the collar, and the button clicks into the spotface



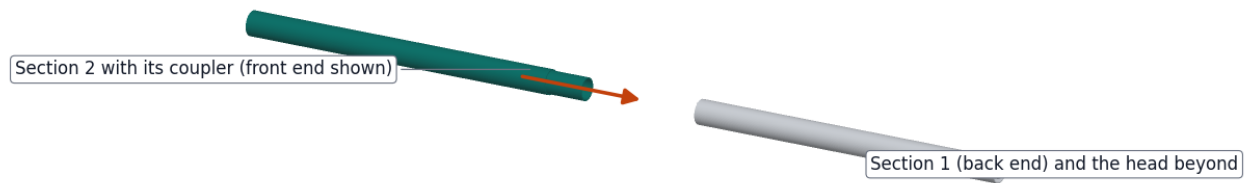
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/voidscope

Slide section 1 forward along the lines, press its front button and push the coupler into the collar socket until the tube butts the collar and the button clicks into the spotface.

Step 9: join sections 2, 3 and 4**Step 9: join sections 2, 3 and 4**

The same at each joint: slide the section along the lines, press its button, push home until it clicks and the tube ends butt



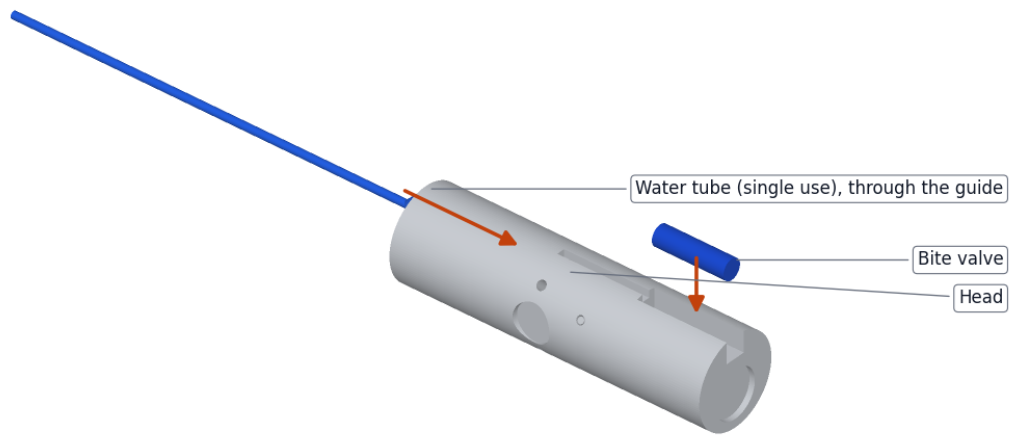
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/voidscope

Slide each section forward along the lines and click it onto the one in front. The depth bands should run on in order. To store the probe, undo the joints and lay the sections side by side in the carry case, with the lines in loose loops at least 150 mm across.

Step 10: water tube through the guide and into the bite valve pocket**Step 10: water tube through the guide and into the bite valve pocket**

Push it in from the surface end until it shows at the exit; lay it in the groove; fit the valve and press it into the pocket



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

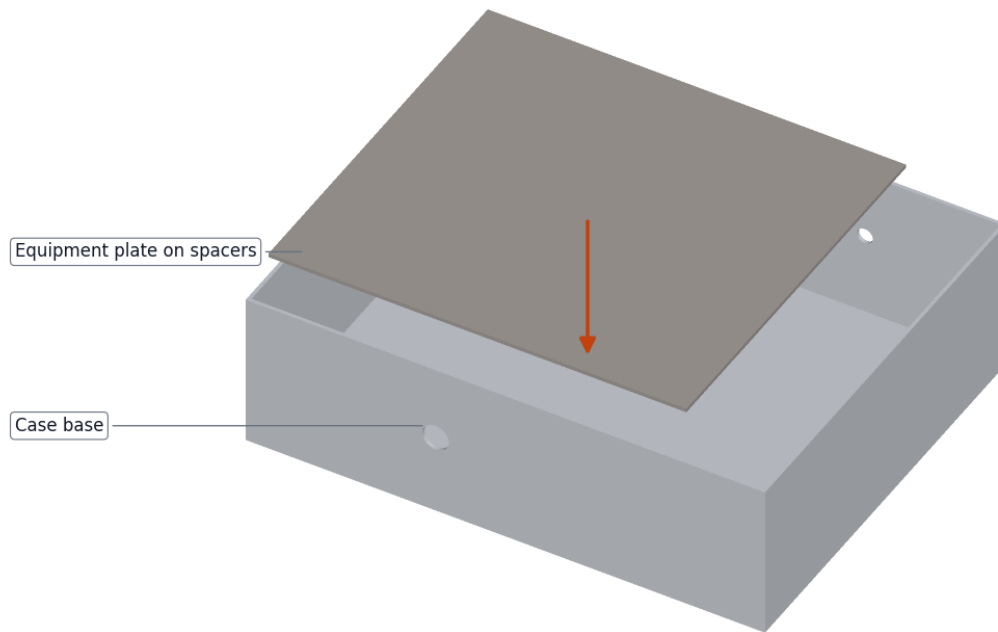
github.com/BoujeeEnjinia1701/voidscope

From the surface end, push a new water tube through the guide tube until it appears in the collar's air chamber; guide it up and out of the 45° exit with a bent wire, lay it in the top groove, fit the bite valve and press the valve into its pocket. Do this with clean gloves; the tube end must touch nothing but the valve.

Step 11: equipment plate into the case

Step 11: equipment plate into the case

Four M5 screws up through the floor with bonded sealing washers, 6 mm spacers, nyloc nuts on top



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

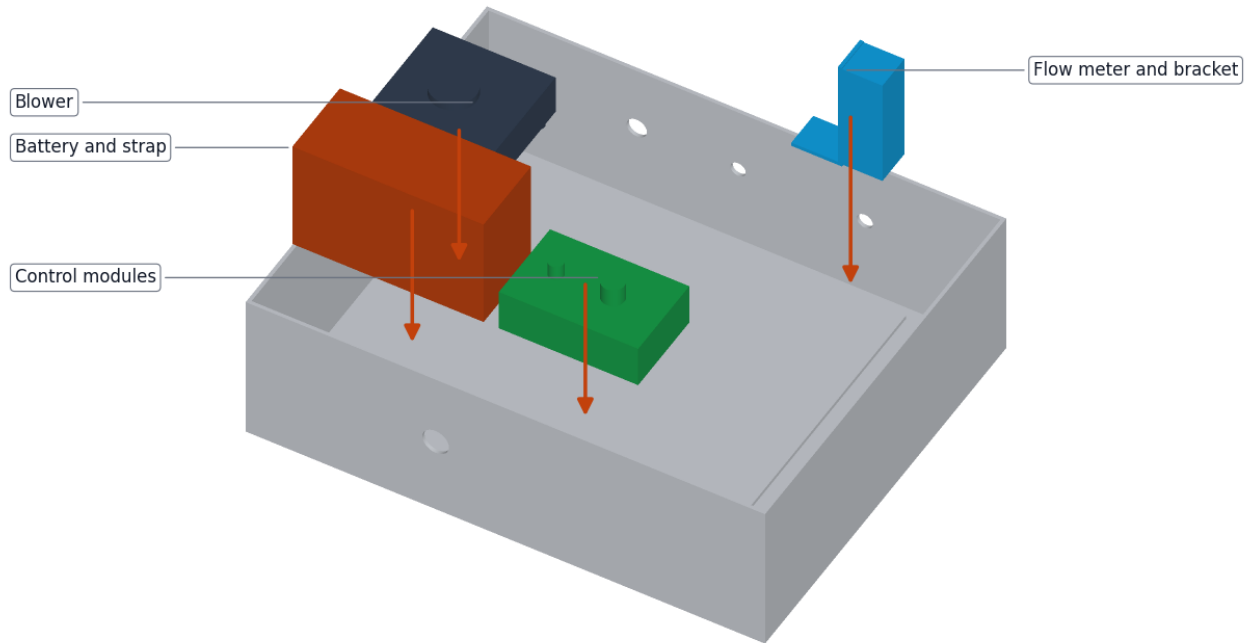
github.com/BoujeeEnjinia1701/voidscope

Four M5 screws up through the floor with bonded sealing washers outside, 6 mm nylon spacers inside, the plate on top, nyloc nuts.

Step 12: battery, blower, modules and flow meter onto the plate

Step 12: battery, blower, modules and flow meter onto the plate

Wire as the wiring picture shows; battery lead unplugged and fuse out



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/voidscope

Fix each part to the plate; strap the battery. Wire them as Figure 18 shows, with the battery lead unplugged and the fuse out. Join the filter, blower, flow meter and outlet with short lengths of hose and hose clips.

VoidScope: wiring, air line and water line (block level)

BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT. All circuits are extra-low voltage: 12.8 V nominal, 14.6 V at most.

Safety: battery lead and fuse out until the stop points in section 6 of the plan are passed.

Red: power. Blue: video. Purple: speaker. Teal: air. The charger is used only away from the kit, on a bench.

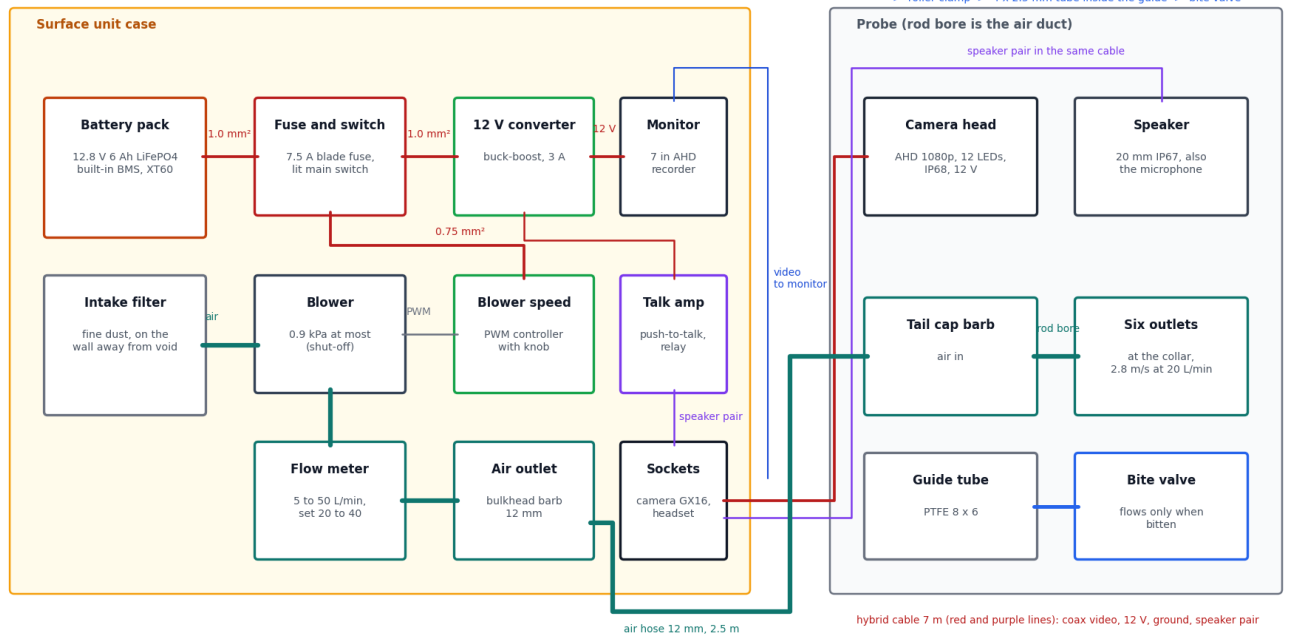
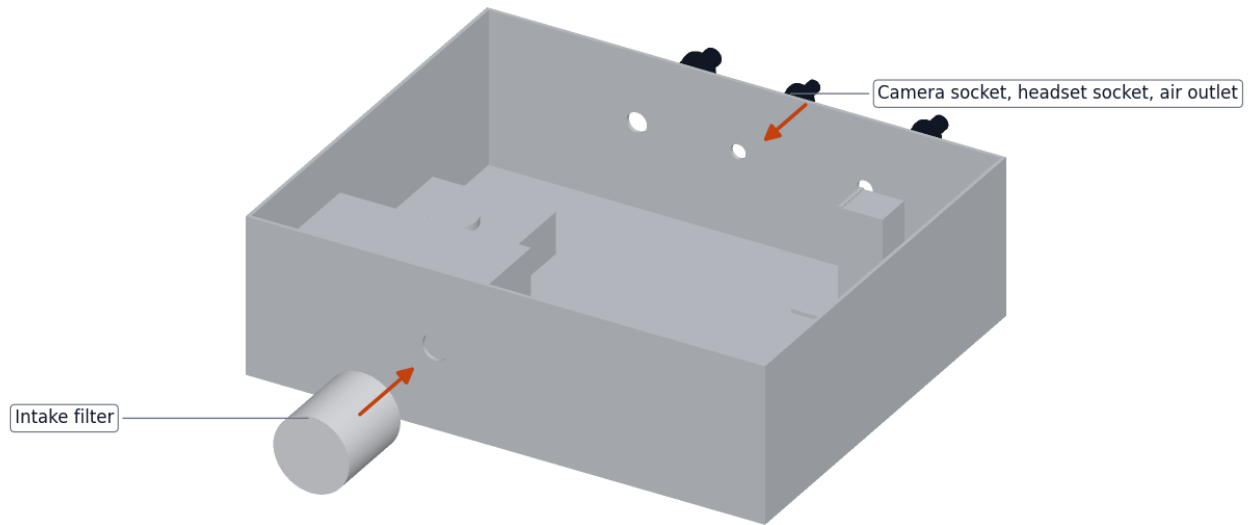


Figure 18. Block-level wiring with wire sizes, and the air and water lines. All circuits are extra-low voltage.

Step 13: wall fittings and intake filter

Step 13: wall fittings and intake filter

Each through its hole with its seal outside and nut inside; hoses from filter to blower to meter to outlet



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

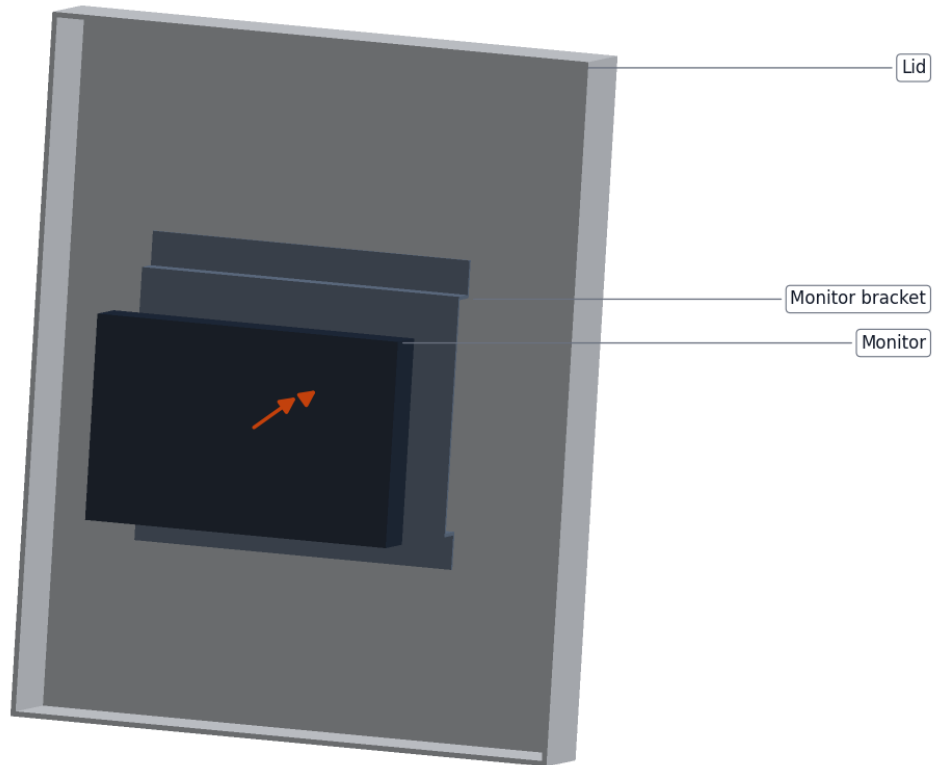
github.com/BoujeeEnjinia1701/voidscope

Fit the camera socket, headset socket and air outlet through the wall facing the probe, and the intake filter through the opposite wall, each with its seal outside and nut inside.

Step 14: monitor and bracket into the lid

Step 14: monitor and bracket into the lid

Bracket feet through the lid on four M5 screws with sealing washers; monitor on its rear M4 holes



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

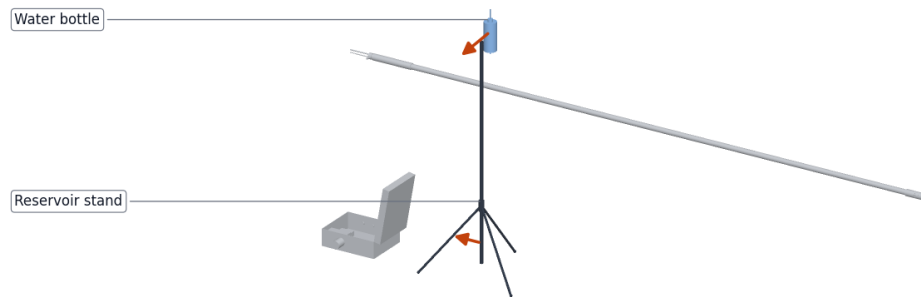
github.com/BoujeeEnjinia1701/voidscope

Bolt the bracket feet through the lid with four M5 screws and sealing washers; screw the monitor to the bracket; run its leads to the modules with enough slack for the lid to open fully.

Step 15: set up at the hole

Step 15: set up at the hole: stand, bottle, hose and plugs

Bottle at most 1 m above the tip; water set to the guide; air hose to the barb; cable plug in; then the safety stops



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/voidscope

Stand the surface unit beside the hole with its lid toward the operator. Plug in the camera cable; join the air hose to the outlet and the tail cap barb; sleeve the hose and lines together. Hang the bottle on the stand no more than 1 m above the tip; join the water set from the bottle through the stopcock, strainer, capillary and roller clamp to the water tube. Then work through the safety stops in section 6.

■ 5. First checks

These are the checks a TRL 4 test report would record; this plan only lists them. Requirement numbers are those of VDS-REQ-001.

Table 2. First checks.

Check	Requirement	How	Pass when
Core hole	R1	Pass the assembled probe through a 51 mm ring along its whole length	It passes without catching at the head, buttons or grip
Reach	R2	Measure from the front of the grip to the tip	4,000 mm or more
Joints	Design loads	Pull each joint by hand, then hang a 30 kg (300 N) load from the tip with the probe vertical; twist each joint	Nothing moves; each button releases with one press
Picture	R3	Dark box, 20 mm letters at 1 m, then with dust	Letters readable on the monitor and in the recording
Sealing	R4	Head under 1 m of water for 30 min with the camera on	Picture stays clear; no water in the head
Water ceiling	R5	Bottle at 1.0 m above the tip, clamp fully open, collect at the bite valve held open for 10 min	Between 1.2 and 1.8 mL/min at room temperature; none with the clamp shut or the valve not bitten
Air	R6	Flow meter at 20 L/min; manometer at a blocked tip at full speed	20 L/min reached; blocked-tip pressure 1.0 kPa or less
Battery	R7	Run video and light from one full pack	4 h or more at room temperature (the -10 °C run is a TRL 4 test)
Setup	R9	Two people from closed cases to a picture	3 min or less
Water set change	R10	Pull the old tube, push a new one, prime	5 min or less, no tools
Voice	R12	Talk with a person at the tip in a noisy yard	Both understood without repeating
Handling	Calculation note B	Push 2 m into a box with the tip unsupported, then 3 m	Held steadily at 2 m; at 3 m the tip must rest on something

■ 6. Safety stops

Stop at each point. Carry on only when everything listed is true.

- **S1. Before the lathe work.** Chuck key out of the chuck; work held firmly (the 6.5 mm offset in a four-jaw chuck is balanced); guards in place; safety glasses on; no gloves at the lathe.
- **S2. Before the battery is plugged in.** The fuse is out; with a meter, there is no short from the battery socket to ground; every wire end is crimped or soldered and covered; the polarity at the converter matches the pack.
- **S3. Before the fuse goes in.** The pack reads 12.8 to 13.6 V, is undamaged and is strapped down; the main switch is off.
- **S4. Before the blower runs on the probe.** With the outlet blocked by a gloved thumb, the blower at full speed cannot be felt pushing hard; a manometer at the outlet reads 1.0 kPa or less. The intake filter is fitted and faces away from any engine or smoke.
- **S5. Before water goes into the line.** The water set is new and sealed until now; the capillary and strainer are in the line; there is no pump, pressurised container or syringe connected to the bottle side; the bottle hangs no

more than 1 m above the tip. Prime with the syringe through the stopcock only before the probe goes in, then turn the stopcock to the bottle and close the syringe port.

- **S6. Before the probe goes into any void, even in training.** The site's structural safety officer has cleared the work position; the operator stays outside the void and off unstable debris; the probe is never used as a lever.
- **S7. Before water is offered to a person (outside this plan; drills only).** The team's medical lead directs it; the person is conscious, can speak and can drink; the flow is set by the roller clamp from closed.
- **S8. Charging.** Packs are charged only on their own charger, away from the kit, on a non-combustible surface, never below 0 °C and never unattended on a first charge.

■ 7. Tools, skills and workspace

Tools. Metal lathe with a three-jaw and a four-jaw chuck, boring bar, parting tool and 1/4 BSP tap (or the machine shop allowance in the bill of materials); drill press with a vee block and a 45° vee block; drills 3 to 14 mm, step drill to 22 mm; small milling machine or a saw and files for the nose pocket and grooves; tube cutter or fine-tooth saw; deburring tool; M4 tap; scribe, straight edge and calipers; sheet folder or vice with hardwood jaws for 2 mm sheet; heat gun; soldering iron, wire strippers and crimpers; multimeter; spirit level; manometer reading to 2 kPa; graduated cylinder (25 mL) and stopwatch; clamp stand.

Skills. Basic turning, drilling and tapping; bonding with structural epoxy; through-hole soldering and crimping; safe handling of lithium iron phosphate packs. No mains wiring is part of the build: the charger is a bought, certified unit.

Workspace. A bench about 2 m long so a section can lie flat; a metalwork corner kept apart from the electronics; a clean area for the water set; the charging spot of S8.

Personal protective equipment. Safety glasses for turning, drilling, cutting and soldering; nitrile gloves for epoxy and for handling the water set; hearing protection when cutting tube; no gloves near a turning chuck or drill.

■ 8. Where the numbers come from

- Model and constructability checks: `cad/src/model.py` (`python cad/src/model.py --check` , 106 checks); STEP and STL exports in `cad/step/` and `cad/stl/` .
- Pictures: `cad/src/build_plan_media.py` , using `.kit/build_views.py` ; written to `docs/05-build-plan/` and `cad/drawings/VDS-DWG-101` to `VDS-DWG-109` .
- General arrangement: `cad/drawings/VDS-DWG-001.pdf` , Rev P2.
- Calculations: `docs/04-calcs/01-sizing.md` (VDS-CAL-001 v0.1) and `docs/04-calcs/sizing.py` .
- Bill of materials: `bom/bom.csv` .
- Decisions: `docs/decisions/0002-design-for-construction.md` (VDS-DDR-002), with VDS-DDR-001; register `docs/06-design-decisions.md` .
- Requirements: `docs/03-requirements.md` (VDS-REQ-001 v0.3).